

Akash Pentakota

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SUMMARY

Data Science and AI enthusiast with hands-on experience building ML models, NLP systems, and computer vision applications. Proficient in Python, Scikit-learn, OpenCV, and deep learning frameworks. Proven ability to deliver production-ready AI solutions with measurable impact.

EDUCATION

Centurion University	Sep 2023 – Present
B.Tech, Computer Science & Machine Learning	GPA: 8.8/10
Narayana Junior College	Apr 2023
MPC (Mathematics, Physics, Chemistry)	GPA: 9.6/10
Vidyadhari EM School	Aug 2021
SSC	GPA: 10/10

EXPERIENCE

Software Development Intern | Oceanautics Private Limited May 2026 – Jun 2026

Python, OpenCV, Computer Vision, Machine Learning, NumPy, Pandas, Git

- Built **Motion Guard AI** — real-time surveillance system using Python and OpenCV; automated motion detection, object tracking, and event monitoring from live video streams.
- Developed **AwakeMate** — AI wellness monitoring platform analyzing behavioral and sleep patterns to deliver personalized recommendations.
- Engineered scalable backend data-processing modules using NumPy, Pandas, and ML libraries, integrated directly into autonomous marine robotics pipelines.
- Managed version control via Git/GitHub; maintained code quality and documentation standards across the team.

PROJECTS

Multimodal Emotion Recognition System  | Nov 2025

Python, Hugging Face Transformers, Scikit-learn, NumPy

- Achieved **87% overall accuracy** by fusing three modalities: BERT text (90%), audio transformer (82%), and CNN/ViT vision (88%).
- Optimized real-time processing of video, audio, and text streams, achieving **15–20 FPS** performance.
- Designed a fusion pipeline that improved reliability by **20%** over single-modality baselines while maintaining inference latency **under 300 ms**.

Motion Guard AI – Real-Time Intelligent Surveillance System  | Jun 2025

Python, OpenCV, YOLOv8, NumPy, Git, GitHub, Computer Vision, AI, ML

- Built a real-time surveillance platform integrating motion detection, object detection, and multi-object tracking using Computer Vision and Deep Learning.
- Developed scalable video analytics pipelines with OpenCV and YOLO for automated monitoring and event detection with low latency.
- Optimized inference workflows through image preprocessing and model tuning, ensuring robust performance across diverse operating conditions.

SKILLS

Languages:	Python, Java, C, SQL
ML / AI:	Scikit-learn, PyTorch, Hugging Face, OpenCV, NumPy, Pandas
Data & Visualizations:	Matplotlib, Power BI, Excel, Google Sheets
Tools:	Git, GitHub, Jupyter Notebook, Eclipse

CERTIFICATIONS

- Applied Data Science (Live) — GeeksforGeeks May 2026
- GenAI Powered Data Analytics Job Simulation — TATA / Forage Feb 2026
- Java Training Program — GeeksforGeeks Oct 2025